

ments, including offices, homes, and retail spaces. Customers with products in the market provide additional data. It is essential to balance power needs based on expected light levels for IoT products using indoor light. We have deployed several products in real-life applications, demonstrating their effectiveness.

How do you help in making ultra-low-power microcontrollers battery-less?

We have seen a convergence between our indoor solar cell technology that can harvest energy in very low light and the decreasing power draw of ultra-low-power microcontrollers. This convergence enables the deployment of low-power IoT sensors indoors. The current challenge is to make these ultra-low-power electronics compatible with solar cells, considering their different behaviour compared to batteries.

Is there any energy storage system? How does the microcontroller operate in the absence of light?

We can deliver around 500 microwatts from our standard 50x50mm PV module in an indoor office environment (500 lux). Microcontrollers consume much less on average but have higher peak currents, so a reservoir is needed. For devices that work only when there is light, a small capacitor is sufficient. If you want the device to operate during the night, weekends, or extended dark periods, a larger storage element like a supercapacitor is needed. For longer durations or seasonal changes, a rechargeable battery, or hybrid capacitor is used. Emerging technologies like printed batteries and solid-state batteries are also being explored.

Does your solution currently include an integrated energy storage system?

We specialise in solar cells and guide IoT companies on adopting

energy harvesting, providing system proposals. While we do not sell complete systems, we offer valuable recommendations. In this collaboration, our contribution is the solar cell. The ultra-low-power microcontroller is specifically designed for energy harvesting, making it a better fit for our solar cells.

What should manufacturers consider when using indoor solar cells in their devices?

Manufacturers in the IoT product space typically have experience with components that are suitable for SMT soldering, reflow soldering, wave soldering, and similar techniques, along with easily accessible connectors. Our product involves thin-film and printed electronics. We have collaborated with the industry to identify effective methods for soldering or gluing components onto a board, as well as integrating mechanical parts seamlessly. Currently, we provide support for these processes. Eventually, these processes may become standard practices, enabling the industry to handle them independently.

How do you ensure the durability of indoor solar cells in varying light conditions?

We conduct extensive long-term accelerated aging tests on our production batches. This involves subjecting the modules to various conditions such as damp climate chambers, ovens, and intense light exposure, both continuously and intermittently. This is done to hasten the exposure of modules to high levels of photons, water molecules, and temperature fluctuations. We also have modules on office walls that have shown consistent and reliable performance over three years.

What does the competitive landscape look like in the indoor solar

cell industry?

For indoor solar cells, it is mainly driven by companies working on organic solar cells, like ours and DSSC, and perovskite solar cells, which show promise but are a bit behind. Some work is going on with gallium arsenide, a high-performance material, but it is costly. Amorphous silicon manufacturers also play a role.

What sets Epishine apart from competitors in the market?

We differentiate through flexible, reliable, high-performance solar cells with a competitive cost structure. We offer both individual cells and series-connected cells, allowing for adaptable voltage output to meet electronic requirements.

How are your indoor solar cells better, and how much more efficient are they?

Our indoor solar cells provide a power conversion efficiency of approximately 15% under 500 lux LED lighting. It is worth noting that there is a new measurement standard for indoor solar cells, and it may take time for the market to adjust. To make fair comparisons, indoor solar cells from various suppliers should be tested under the same lighting conditions.

What other applications do you see for indoor solar cells besides IoT?

Indoor solar cells find use in remote controls, electronic shelf labels, and low-power consumer electronics such as earphones, keyboards, and mice. While we have not launched products in this market, we are aware of competitors who have. We are actively discussing partnerships and opportunities in various sectors. Currently, our main focus is on IoT connected devices, but we are also exploring other markets, with a keen interest in electronic shelf labels. **EFY**